Predicting and Profiling Potential Caravan Policyholders: A Machine Learning Approach on Dutch Customer Data

Bigoni Beatrice Student ID: 879938
Cavallo Davide Student ID: 880593
Rossini Luca Student ID: 890887

Abstract. In the modern digital economy, data has emerged as a strategic asset, comparable in value to traditional commodities like gold or oil. While the accumulation of data is widespread, the primary challenge lies in extracting actionable intelligence from this raw information. Machine Learning provides the framework to address this, converting data into specific business solutions.
This is particularly relevant in the insurance sector, which frequently relies on direct mail campaigns to market products. Although potentially effective, these mass solicitations—often perceived as "junk mail"—frequently end up discarded, resulting in significant financial waste.
Therefore, gaining a precise understanding of the customer base is essential to minimize these costs and focus resources on the right people.

This report analyzes a dataset comprising sociodemographic and product ownership variables to identify potential customers for caravan insurance policies. By applying advanced classification techniques, we aim to isolate the most promising profiles for targeted and efficient marketing campaigns.

1 Introduction

The Dutch data mining company *Sentient Machine Research* has collected data on thousands of customers.
Due to privacy and geomarketing considerations, sociodemographic attributes have not been measured at the individual level, but rather aggregated based on the postal code of residence. Specifically, the dataset includes demographic statistics, the number of insurance policies held, and the economic contribution of these policies within the area where the individual lives.

This dataset serves as the foundation for this report. Through the application of Machine Learning techniques, the data has been analyzed to address the following **research questions**:

- **RQ1:** *Is it possible to accurately predict which customers are interested in purchasing a caravan insurance policy?*
- **RQ2:** *Which sociodemographic characteristics are the key drivers influencing this purchasing decision?*

2 About the Dataset

One of the primary challenges of this project was managing the high dimensionality of the data. The dataset comprises a total of **87 variables**, structured as follows:

- **1 Metadata Variable:** Specifies the origin of the record, indicating whether the instance belongs to the training set or the test set.
- **43 Socio-demographic Variables:** These attributes describe statistical characteristics of the customer's postal code area (e.g., income, religion, household size). They are identified by the prefix **"M"** at the beginning of the variable name (e.g., *MOSTYPE*).
- **21 Product Contribution Variables:** These variables refer to the financial contribution of specific insurance policies held by the customer. They are identified by the prefix **"P"** (e.g., *PPERSAUT* for car policy contribution).
- **21 Product Count Variables:** These variables report the number of policies held for each specific insurance category (corresponding to the same categories as the "P" variables). They are identified by the prefix **"A"** (e.g., *APERSAUT*).
- **1 Target Variable (CARAVAN):** This is the response variable (y) for our classification task. It is a binary variable where **1** indicates that the customer purchased a caravan insurance policy, and **0** indicates otherwise.

2.1 Variable Types and Encoding

Almost all attributes in the dataset are categorical, with two notable exceptions. The variables *MAANTHUI* (Number of houses) and *MGEMOMV* (Average size of household) are the only **discrete quantitative** variables, accepting integer values ranging from 1 to 10 and 1 to 6, respectively.
The remaining variables are categorical. While most follow an **ordinal** scale (where the order of values has meaning, such as age groups or income ranges), specific attributes use distinct encoding dictionaries:

- **MOSTYPE** (Customer Subtype): Defined by the dictionary **L0**. It consists of 41 **nominal** (non-ordinal) categories representing detailed customer profiles (e.g., "High Income, expensive child", "Very Important Provincials", "Mixed rurals").
- **MOSHOOFD** (Customer Main Type): Encoded using dictionary **L2**, containing 10 nominal categories.
- **MGEMLEEF** (Average Age): Encoded using the ordinal dictionary **L1**.
- **Other "M" Variables:** The remaining sociodemographic variables are encoded using the ordinal dictionary **L3** (percentage ranges).
- **"P" and "A" Variables:** All product ownership and contribution variables are encoded using the ordinal dictionary **L4**.

The specific decoding keys for these dictionaries are provided in the tables below.

Table 1: Dictionary L1: Average Age

Key	Description
1	20–30 years
2	30–40 years
3	40–50 years
4	50–60 years
5	60–70 years
6	70–80 years

Table 2: Dictionary L2: Customer Main Type

Key	Description
1	Successful Hedonists
2	Driven Growers
3	Average Family
4	Career Loners
5	Living Well
6	Cruising Seniors
7	Retired and Religious
8	Family with Grown-ups
9	Conservative Families
10	Farmers

Table 3: Dictionary L3: Percentages

Key	Range
0	0%
1	1 – 10%
2	11 – 23%
3	24 – 36%
4	37 – 49%
5	50 – 62%
6	63 – 75%
7	76 – 88%
8	89 – 99%
9	100%

Table 4: Dictionary L4: Contribution/Counts

Key	Range / Amount
0	0
1	1 – 49
2	50 – 99
3	100 – 199
4	200 – 499
5	500 – 999
6	1000 – 4999
7	5000 – 9999
8	10,000 – 19,999
9	$\geq$ 20,000

2.2 Data Exploration

The data collection process involved measuring these attributes for a total of **9822** customers.
A preliminary quality assessment revealed that the dataset is robust and complete, with **no missing values** recorded across any of the instances.

However, the distribution of the target variable *CARAVAN* introduces a significant challenge: the dataset is severely imbalanced.
As is typical in insurance contexts, the positive class is rare, with only **6%** of customers owning a caravan policy. This imbalance must be carefully addressed, as standard machine learning algorithms tend to bias towards the majority class.

3 RQ1: Classification

3.1 Data Preprocessing

Building upon the insights derived from the Data Exploration phase, two critical challenges were identified: the high dimensionality of the feature space and the severe imbalance of the target class distribution. Consequently, the **Data Preprocessing** strategy was structured to specifically address these issues through two primary phases:

- **Feature Selection**
- **Handling Class Imbalance**

3.1.1 Feature Selection

Machine learning algorithms are sensitive to the dimensionality of the input data, as an increasing number of features leads to an exponential growth of the feature space, giving rise to the Curse of Dimensionality.
To mitigate this issue, **Feature Selection** was applied. This crucial step aims to reduce dimensionality by identifying and removing irrelevant or redundant attributes, thereby simplifying the learning process and enhancing the generalization capability of the models.
Given the predominance of categorical attributes in the dataset, feature extraction techniques such as **Principal Component Analysis (PCA)** were excluded, as they are unsuitable for non-continuous data structures. Consequently, the dimensionality reduction process relied on the following two feature selection approaches:

- **Filter Methods**
- **Embedded Methods**

Filter Methods
Filter methods assess the relevance of features by analyzing the intrinsic properties of the data. In this study, these methods were utilized to address the issue of **multicollinearity**, a condition where two or more predictor variables are highly correlated. To achieve this, a **correlation-based** approach was employed.
Specifically, a **Linear Correlation Matrix** was computed on the training dataset. The rationale behind this step is rooted in the concept of **information redundancy**: if two features exhibit a very strong linear relationship, they essentially convey the same information to the model. Consequently, retaining both contributes to computational inefficiency without enhancing predictive power.

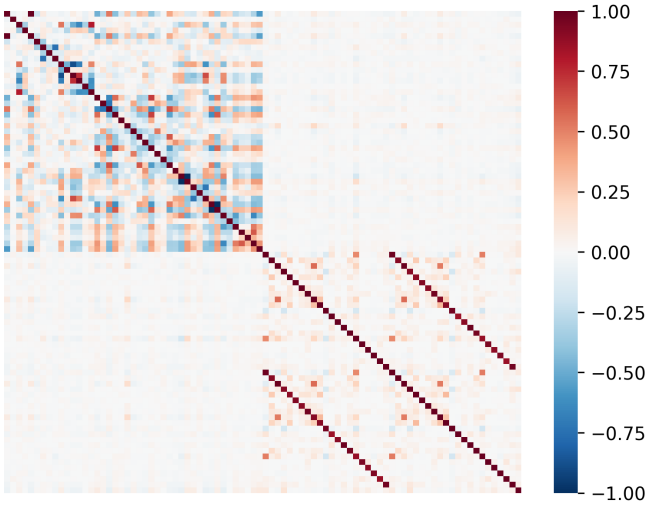


Figure 1: Heatmap of correlation matrix

As illustrated in Figure 1, distinct blocks of high correlation are evident within the dataset.

Consequently, by applying a quantitative threshold of $|\rho| > 0.85$, the following specific redundancies were isolated:

- **Product Variables (P vs. A):** A systematic correlation was observed between all *Product Contribution* variables (prefix 'P') and their corresponding *Product Count* variables (prefix 'A'). This is statistically inherent, as the monetary contribution is directly tied to the number of policies held.
- **Demographic Variables (M):** Three specific pairs of sociodemographic attributes exhibited strong inverse correlations:
  - *Marital Status:* $MRELGE$ (Married) vs. $MRELOV$ (Other relationships);
  - *Housing:* $MHHUR$ (Rented house) vs. $MHKOOP$ (Home owners);
  - *Health Insurance:* $MZFONDS$ (National Health Service) vs. $MZPART$ (Private Health Service).

Consequently, to minimize redundancy, one variable from each identified pair was discarded. Through this pruning process, the dimensionality of the feature space was reduced from the initial **85 predictor variables** to **61**.

Embedded Method
To complement the filter approach, an **Embedded Method** based on the **Random Forest** algorithm was implemented. In this configuration, feature relevance was assessed by analyzing the hierarchical structure of the generated trees.
Particular emphasis was placed on the attributes utilized at **Split Level 0** (root nodes). Since variables selected for the root split maximize the information gain at the very beginning of the decision process, they are considered the most discriminative.
Consequently, based on the usage frequency at this top hierarchical level, the **top 20 variables** were identified and selected for evaluation.

Following a comparative empirical analysis, it was observed that classifiers trained on the **61 variables** (resulting from the correlation filter) yielded performance metrics comparable to those utilizing only the top **20 attributes** identified by the embedded method.
This substantial similarity indicates that the additional 41 variables did not introduce detrimental noise or overfitting to the model. Consequently, the broader set of 61 features was retained for the subsequent analysis to avoid potential information loss from an overly aggressive reduction.

3.1.2 Feature Encoding Strategy

The dataset contains a significant number of variables that possess an intrinsic order. Nevertheless, the distance between levels is not necessarily linear (e.g., the socioeconomic difference between class 0 and 1 may not be

equivalent to the difference between 8 and 9).
To address this, a **Categorical Encoding approach** has been adopted. Most nominal and ordinal variables have been converted into **string format** to treat them as distinct categories rather than continuous values.

3.1.3 Handling Class Imbalance

Imbalanced datasets pose a significant challenge for machine learning classifiers. Standard algorithms tend to be biased toward the majority class to maximize overall accuracy, often leading to the misclassification of minority samples or treating them as noise.
To address this issue, two complementary strategies were explored:

- **Oversampling and Undersampling**
- **Cost Sensitive Learning**

In addition to the class imbalance challenge, the training dataset was affected by the presence of numerous **inconsistent instances**. These are defined as records sharing identical values for all predictor attributes but assigned to different response variable classes (y).

Data Polishing: Handling Inconsistencies
Such contradictions can severely hinder the learning process, preventing the classifier from establishing clear decision boundaries. To address this issue, a **data polishing** strategy was implemented on the training set.
Specifically, all inconsistent instances—defined as records possessing identical feature vectors (attributes) but conflicting target labels—were identified and removed.
This process was executed prior to addressing the class imbalance problem. This chronological order is critical, particularly when employing **oversampling** techniques.
In fact, performing oversampling on an unpolished dataset would inevitably replicate the inconsistent instances.

3.2 Model Selection

Prior to the model selection phase, the dataset was partitioned into two independent subsets: a **Training Set** (75% of the data) used for model learning and validation, and a **Test Set** (25% of the data) reserved exclusively for the final performance assessment.

To address the class imbalance within the Training Set, the experimental framework was structured around two distinct methodological approaches applied to the training data:

- **Data-Level Approach:** The training data was rebalanced using resampling techniques (Oversampling and Undersampling) prior to fitting the models.
- **Algorithmic-Level Approach:** The original training distribution was retained, and the imbalance was managed directly by the algorithms using a **Cost Matrix** (Cost-Sensitive Learning).

3.2.1 Approach I: Resampling Techniques (Oversampling and Undersampling)

Specifically, the classifiers were trained on two separate datasets: the expanded dataset resulting from **Oversampling** (comprising 13,202 instances) and the reduced dataset resulting from **Undersampling** (comprising 728 instances).

To identify the most suitable predictor, five classification algorithms were tested. These models were specifically selected for their inherent capability to handle categorical variables and non-linear relationships effectively, avoiding the methodological limitations associated with distance-based algorithms (such as KNN or SVM) when dealing with high-dimensional nominal data. The tested models are:

- **Decision Tree (without pruning)**
- **Decision Tree (pruned)**
- **Random Forest**
- **Naive Bayes**
- **XGBoost**

To ensure the robustness of the results and minimize the variance in performance estimation, every model was evaluated using **Stratified 10-Fold Cross-Validation**.

Results from Oversampling and Undersampling
Following the Cross-Validation phase, the performance of the five classifiers was analyzed using Receiver Operating Characteristic (ROC) curves.

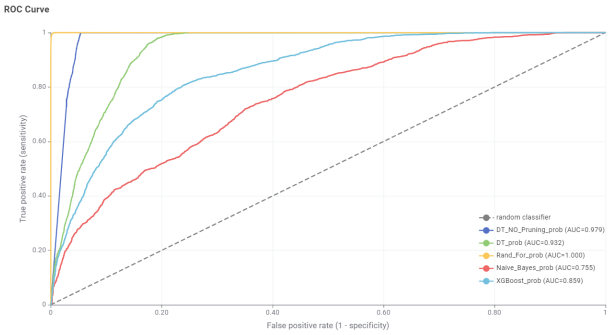


Figure 2: ROC curves for models trained after Oversampling

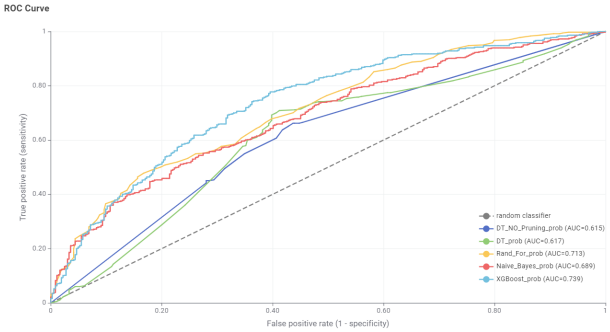


Figure 3: ROC curves for models trained after Undersampling

As illustrated in Figure 2, the classification results obtained from the **oversampled training set** appear exceptionally high. The Random Forest classifier achieved an Area Under the Curve (AUC) of 1.0, representing a theoretically perfect classification. Similarly, the Decision Tree models exhibited extremely high metrics, with Accuracy ranging between 0.888 and 0.972, and Cohen's Kappa between 0.778 and 0.945. Conversely, Naive Bayes and XGBoost showed significantly lower performance compared to the tree-based methods, with Accuracy values of 0.68 and 0.491, and Cohen's Kappa of 0.36 and 0.012, respectively.

On the other hand, the results derived from the **undersampled training set** (Figure 3) are notably more conservative. In this scenario, the performance gap between the different classifiers is less pronounced. It is worth noting that XGBoost and the Pruned Decision Tree recorded Accuracy values close to 0.5, a phenomenon that requires further interpretation.

Discussion and Interpretation of Results
The discrepancy between the two experimental setups necessitates a critical analysis of the underlying causes.

Regarding the **Oversampling** results, the near-perfect metrics strongly suggest a case of **Data Leakage**.
The strategy employed involved **Random Oversampling** (via replication), which creates exact copies of the existing minority records to balance the class distribution. Due to this duplication, identical instances were present throughout the dataset. Consequently, during the cross-validation partitioning, these duplicates were distributed across both the training and validation folds, and this overlap allowed the model to simply "recognize" the specific records it had already encountered during training, rather than learning generalizable rules.
This explains the inflated AUC of 1.0, which reflects overfitting driven by data redundancy rather than genuine predictive capability.

Conversely, the results from the **Undersampling** strategy are considered more realistic, despite the lower Accuracy figures.

The observed divergence between high AUC scores (e.g., for XGBoost) and low Accuracy (≈ 0.5) is attributed to a **Probability Calibration Shift**. The models were trained on a perfectly balanced dataset ($P(y = 1) = 0.5$), leading them to assign higher posterior probabilities to the minority class. When applied to validation folds that reflect the natural imbalance, the standard decision threshold of 0.5 results in a high number of False Positives.

3.2.2 Approach II: Algorithmic-Level Approach (Cost-Sensitive Learning)

While the Data-Level resampling techniques discussed previously provided an initial mitigation strategy for class imbalance, an Algorithmic-Level approach was subsequently adopted to enhance model robustness without artificially manipulating the underlying data distribution. Consequently, **Cost-Sensitive Learning algorithms** were evaluated.
Two distinct methodologies were applied: **Direct Approaches** (modifying the learning algorithm's loss function) and **Meta-learning Approaches** (modifying the prediction threshold during post-processing).

Direct Approaches on XGBoost
This methodology involves directly integrating the misclassification cost into the training procedure of the classifier (Intrinsic Cost-Sensitivity).
Specifically, the **XGBoost** algorithm was selected for this stage due to its capability to accept a `scale_pos_weight` parameter. To ensure consistency with the previous Data-Level approach, a **K-fold Cross-Validation** scheme was maintained.
Four distinct configurations were tested:

- **XGBoost Standard (Weight 1):** Used as a baseline for performance comparison.
- **XGBoost with Weight 2 & 3:** Tested as heuristic incremental penalizations.
- **XGBoost with Weight 15:** Calculated based on the exact **Imbalance Ratio** of the dataset (Negative/Positive $\approx 93/6 \approx 15.5$).

By assigning a higher weight, the algorithm penalizes False Negatives (missed Caravan buyers) significantly more than False Positives. This forces the model to prioritize the minority class during the gradient boosting process, unlike the standard unweighted model.

Meta-learning Approaches (Post-processing)
For classifiers lacking intrinsic cost-sensitive parameters (such as Random Forest, Naive Bayes, and Decision Tree), a methodology based on **post-processing** was utilized. This approach modifies the outputs of the classifier during the inference phase rather than the training phase.

The models were evaluated using the same **K-fold Cross-Validation** strategy employed previously.

However, instead of relying on the default classification threshold ($\tau = 0.5$), the decision boundary was optimized using the *Binary Classification Inspector* node in KNIME. This allowed for the calibration of the threshold to minimize the expected cost, effectively compensating for the model's bias towards the majority class.

Results from XGBoost

The comparative analysis highlights the efficacy of the weighting strategy. The results obtained from the XGBoost weighted 15 and XGBoost weighted 3 are displayed below.



Figure 4: Display from Binary Classification Inspector for XGBoost weighted 15



Figure 5: Display from Binary Classification Inspector for XGBoost weighted 3

As illustrated in Figure 4, the **Sensitivity (Recall)** exhibits a significant improvement, rising from a negligible 0.05 (in lower-weighted models) to **0.66** with Weight 15.

Although this increase in Recall is accompanied by a reduction in Accuracy, this trade-off is aligned with the business objective. In the context of caravan insurance marketing, failing to identify a potential customer (False Negative) represents a significant missed revenue opportunity, whereas contacting a disinterested customer (False Positive) incurs only a minor operational cost.

Furthermore, by performing **Threshold Tuning** on the weighted model, in particular lowering the decision threshold from 0.5 to **0.3**, it was possible to classify as positive any instance with a probability greater than 30%. This adjustment further maximized the Sensitivity to **0.88**, ensuring the capture of the vast majority of potential policyholders.

Results from Meta-learning Approaches

The post-processing strategy, based on threshold optimization, was applied to the classifiers lacking intrinsic cost-sensitivity (Decision Tree, Random Forest, and Naive Bayes). The analysis of the results indicates that, unlike the Direct Approach used with XGBoost, this methodology yields limited improvements in addressing the underlying class imbalance.

As evidenced by the performance of the Decision Tree (Figure 6), which is representative of the behavior observed across the other standard classifiers, the model remains heavily biased towards the majority class. The predicted probability distribution for the positive class is strongly skewed towards zero, mirroring the prior probability of the dataset ($\approx 6\%$).

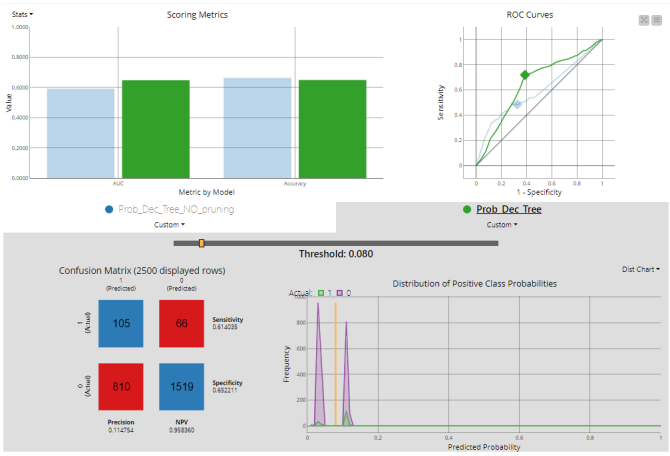


Figure 6: Display from Binary Classification Inspector for Decision Tree: Probability Distribution and ROC Curve

It was observed that to achieve a significant level of Sensitivity (e.g., 0.61), the decision threshold must be drastically lowered to extremely conservative values ($\tau \approx 0.08$).

Furthermore, the ROC curve exhibits a stepped structure ("spiky" behavior), suggesting poor probability calibration and a limited ability of the standard algorithms to distinguish the minority class without intrinsic weighting.

Consequently, while the Meta-learning approach effectively forces the retrieval of potential customers by accepting a higher rate of False Positives, it does not fundamentally resolve the learning bias. The models continue to classify the vast majority of instances with low probabilities, necessitating aggressive and potentially

unstable threshold adjustments to be operationally useful.

3.3 Final Evaluation on Test Set

The best-performing models derived from the **Data-Level (Resampling)** and **Algorithmic-Level (Cost-Sensitive)** approaches were evaluated on the held-out Test Set to verify generalization capabilities. The results highlight a significant performance divergence between the two strategies:

- **Data-Level Approaches (Decision Tree & Random Forest):** A distinction must be made between the two resampling techniques employed. The **Oversampling** method yielded the highest Accuracy, reaching approximately 0.90. However, this result suggests potential issues of data leakage or overfitting, likely caused by the exact duplication of minority class records during training. The **Undersampling** method also maintained a respectable Accuracy of approximately 0.67. Nevertheless, a critical limitation was observed in both approaches: the **Sensitivity (Recall)** for the minority class remained critically low at approximately 0.10, indicating an inability to effectively identify target customers on new data.

- **Algorithmic-Level Approach (XGBoost Weight 15):** The Cost-Sensitive XGBoost demonstrated a more robust and balanced performance. The model achieved an optimal Accuracy of approximately 0.74. Most importantly, it recorded the highest **Sensitivity** among all tested models on the Test Set, reaching a value of **0.66**.

3.3.1 Conclusion

Based on the comparative analysis, the **Weighted XGBoost** is identified as the superior solution for the sales strategy. By prioritizing the retrieval of potential customers (maximizing Recall) while maintaining stable accuracy, this model aligns best with the business objective of market penetration. Consequently, it is selected as the final champion model.

4 RQ2: Clustering Analysis

In this phase, **unsupervised learning techniques** were employed to partition the customer base into distinct segments based on their sociodemographic characteristics. The primary objective was to unveil latent patterns and identify homogeneous profiles without relying on the target variable (`CARAVAN`).
In order to apply this process it has been adopted two different algortihms:

- **K-Medoids**

- **K-Means**

4.1 Data Pre-processing

4.1.1 Normalization

To ensure the reliability of the algorithms, which rely on Euclidean distance, it has been applied a **Min-Max Normalization** to all numerical variables prior to clustering.

4.1.2 Feature engineering on categorical variables

Prior to modeling, categorical variables were handled via **One-Hot Encoding**.

To address the high dimensionality of the `MOSTYPE` feature, a **Domain Knowledge Feature Engineering** technique was applied.
As detailed in **Table 5**, the 41 subtypes were aggregated via a *Rule Engine* into 5 logical macro-categories (e.g., "Affluent", "Families", "Elderly"). These new categories were dummy-encoded and included in the K-Means model alongside other normalized demographic variables.

Table 5: Feature Engineering: Aggregation Rules for Customer Segmentation

Original MOSTYPE Codes	**Mapped Segment**
1, 2, 3, 4, 12, 14, 15, 23	**Affluent & High-Status**
6, 8, 9, 10, 11, 13, 34, 36, 38	**Families & Middle-Class**
7, 16, 17, 18, 21, 22	**Urban Singles & Young**
5, 25, 26, 27, 28, 29, 30, 32	**Elderly & Retirement**
19, 20, 24, 31, 33, 35, 37, 39, 40, 41	**Rural & Low-Income**
Any other value	*Other*

4.2 Feature selection

In all methodologies, the variables related to policy ownership (`P`) and contributions (`A`) were preliminarily removed.
This choice was dictated by the need to profile customers exclusively based on sociodemographic characteristics (`M`), avoiding clustering influenced by the existing product portfolio, which would have generated groups based on past behavior rather than future potential.

To ensure the robustness of the segmentation, the K-Means and K-Medoids algorithms were evaluated under two distinct configurations:

- **Scenario A:** Including all available attributes, comprising the socio-demographic variables `MOSTYPE` and `MOSHOOFD`.

- **Scenario B:** Excluding `MOSTYPE` and `MOSHOOFD` to assess clustering performance based solely on behavioral and ownership data.

4.3 Comparative Analysis of Clustering Models

The models were compared based on two conflicting metrics:

- **Geometric Cohesion (Silhouette Coefficient):** Measures how well-separated and compact the clusters are mathematically (range -1 to 1).
- **Business Utility (Conversion Rate):** Measures the percentage of policy buyers (`CARAVAN`=1) within the best-performing cluster, compared to the global average of 6%.

4.3.1 Summary of Results

Table 6 summarizes the performance comparison. A stark contrast emerged between the two scenarios: the inclusion of aggregated sociodemographic variables led to a collapse in cluster separation, whereas their removal allowed actionable patterns to emerge.

Table 6: Clustering Results Comparison (Global Avg: 6%)

Algorithm	Scenario	Silhouette	Best Cluster CR
K-Means	A (With Vars)	0.00	10.4%
K-Medoids	A (With Vars)	≈ 0.00	8.9%
K-Means	B (No Vars)	0.09	10.5%
K-Medoids	**B (No Vars)**	**0.067**	**12.1%**

4.3.2 Discussion: Geometric Noise vs. Business Utility

The analysis highlighted how redundant variables can degrade clustering performance rather than enhance it.

Scenario A: Contrary to expectations, including the aggregated variables (`MOSTYPE`, `MOSHOOFD`) resulted in a Silhouette score of **0.00** for both algorithms. This indicates a total overlapping of clusters with no distinct geometric separation. **Implication:** This confirms that adding these derived sociodemographic labels introduced excessive "noise" into the distance matrix, preventing the algorithms from detecting any clear boundaries between groups.

Scenario B: By removing the redundant variables, the cluster structure began to recover. While the Silhouette scores remained low ($0.067 - 0.09$)—which is typical for high-dimensional behavioral data—they improved significantly compared to Scenario A. Crucially, this "cleaner" environment allowed the **K-Medoids** algorithm to detect a latent niche that was previously masked, isolating a specific segment with a **12.1% Conversion Rate**.

4.4 Conclusion and Model Selection

The **K-Medoids (Scenario B)** is selected as the optimal solution.

Although its Silhouette score (0.067) indicates fuzzy boundaries between groups, it is the only model capable of delivering a high business impact. By prioritizing **Lift** over geometric perfection, this model successfully identified a target group where the probability of purchasing a policy is **double the global average** (12.1% vs 6%).

This result justifies the exclusion of the sociodemographic variables, proving that in this context, behavioral data holds greater predictive power than static social labels.

5 Conclusion

The classification and clustering tasks performed on the CARAVAN dataset presented significant challenges, primarily due to the inherent complexity of the insurance domain. The analysis confirmed that consumer behavior in this sector is not a deterministic science; the decision to purchase a caravan policy is influenced by multifactorial and often latent drivers that cannot be perfectly captured by sociodemographic attributes alone.
Consequently, obtaining "perfect" classifiers with zero error rates was not methodologically feasible given the stochastic nature of the real-world data.
Similarly, the unsupervised segmentation phase revealed that customer profiles do not naturally organize into geometrically distinct groups. The low Silhouette coefficients observed across all models indicate that the boundaries between different customer types are nuanced and overlapping rather than sharply defined.

Despite these structural limitations, the dataset was leveraged to its fullest potential through a rigorous methodological approach. In the classification phase, advanced techniques for handling class imbalance were adopted, prioritizing business-centric metrics (such as Profit and Potential) over standard accuracy. In the clustering phase, notwithstanding the geometric "fuzziness," the analysis successfully isolated a high-value segment where the propensity to purchase is approximately **double the global average** (12.1% vs 6%).

In conclusion, while the resulting models reflect the noise and unpredictability of human behavior, they successfully extract actionable intelligence, providing the marketing department with specific, high-probability targets to optimize campaign efficiency.

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